Total No. of Questions : 8 | | Total No. of Printed Pages : 4

Paper Code : 21307

F-407

B.C.A. (Second Semester)

Examination, 2024

(New Course)

Paper No. BCA-N-202

DISCRETE MATHEMATICS

Time : Three Hours] *[Maximum Marks : 70*

Note : Attempt any **five** questions. **All** questions carry equal marks.

1. (a) Differentiate between conjunction and disjunction. Provide an example for each.

   (b) Discuss the duality of conditional and unconditional statements.

2. (a) Prove that if a graph (connected or diconnented) has exactly two vertices of odd degree, then there exist a path joining these two vertices.

   (b) Prove that every connected graph with n vertices and n−1 edges is a tree.

3. (a) Prove that a Boolean algebra is a self dual.

   (b) Show that the union of two equivalence relation need not be an equivalence relation.

4. (a) Show that:

   (a) $\sim(p \vee q) \vee (\sim p \wedge q) \vee p$ is a tautology

   (b) $((p \vee q) \wedge \sim q) = p$ is a tautology

   (c) $(p \vee q) \wedge (\sim p \wedge \sim q)$ is a contradication.

   (b) Minimize the four variable logic function using K-Map

   f(A, B, C, D) = A B C'D + A'BCD + A'B'C' + A'B'D' + AC' + AB'C + B'

5. (a) Simplify the expression (x+y) (x+z) using the laws of boolean algebra.

   (b) Using Boolean identities, reduce the given Boolean expression:

   F(X, Y, Z)=X'Y+ YZ' + YZ + XY'Z'

6. (a) Let A= {0, 1, 2, 3, 4} and define a relation R on A as follows:

R={(0, 0), (0, 4), (1, 1) (1, 3), (2, 2), (3, 1), (3, 3), (4, 0), (4, 4)}.

R is an equivalence relation on A. Find the distinct equivalence classes of R.

(b) Prove that the following argument is valid:

"all dogs are carnivorous."

"some animals are dogs."

Therefore" some animals are carnivorous".

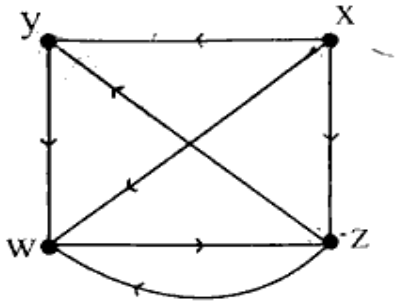
7. (a) Construct the binary tree whose inorder and preorder traversal is given below.

Also, find the postorder traversal of the tree.

Inorder : d, g, b, e, i, h, j, a, c, f

Preorder : a, b, d, g, e, h, i, j, c, f

(b) Consider the directed graphy G in figure:



(a) Find indegree and out degree of each vertex of G.

(b) Find all simple paths from y to z

(c) Find all cycles in G.

8. (a) Use truth table to prove the following argument

$P \rightarrow \sim q$

$r \rightarrow q$

$\frac{q}{\therefore \sim r}$

(b) Let $g: A \rightarrow B$ and $f: B \rightarrow C$ be two functions. Show that if g and f are both injectives, then $f \circ g : A \rightarrow C$ is injective.

Total No. of Questions : 8 | | Total No. of Printed Pages : 3

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B.C.A. (Second Semester) Examination, 2023

New Course

Paper No. BCA-N-202

DISCRETE MATHEMATICS

Time : Three Hours] *[Maximum Marks : 70*

Note : Attempt any **five** questions. **All** questions carry equal marks.

1. (a) Explain proposition and propositional form conditional.

   (b) Explain conjunction and disjunction with example.

2. (a) Simplify the following Boolean function

   (i) $\overline{A}B\overline{C} + ABC + \overline{A}BC + AB\overline{C}$

   (ii) $\overline{A}B + AB + B\overline{C}$

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   (b) Draw k-map

   $F(A,B,C) = \Sigma(2,3,4,6,7)$

3. (a) Give an example of a graph that is regular, but not complete, with each vertex having degree three.

   (b) Explain trees and its properties.

4. (a) Let $f:A \rightarrow B$ and $g:B \rightarrow C$ be functions. Show that if gof is onto then g is onto.

   (b) If A has m elements and B has n elements, how many functions are there from A to B?

5. (a) Explain min-terms and max-terms with example.

   (b) Explain AND, OR NOT operators.

6. (a) Prove that relation R defined on any non-void set A as $(a,b) \in R \Leftrightarrow a \geq b$ is partial order relation.

   (b) Explain equivalence relation with example.

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7. (a) What is graph? Explain finite and infinite graphs with example.

(b) Explain incidence matrix with example.

8. Write short notes on any **four** of the following :

(i) Boolean postulates

(ii) De Morgan's theorem

(iii) Connected graphs

(iv) Truth tables

(v) Inverse function

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B. C. A. (Second Semester) EXAMINATION, 2019

(New Course)

Paper No. BCA—N– 202

DISCRETE MATHEMATICS

Time : Three Hours] [*Maximum Marks : 70*

Note : Attempt any *five* questions. All questions carry equal marks.

1. (a) Explain duality conditional and in-conditional statements.

   (b) Explain conditional and bi-conditional operators, with an example.

2. Simplify the following Boolean function :

$$F(A, B, C, D) = \Sigma(0, 1, 2, 8, 9, 10)$$

in :

(i) Sum of products form

(ii) Product of sum form

(iii) Give implementation of the function using gates.

(iv) Construct truth table also.

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3. (a) Define the terms with respect to graph : Graph, path, cycle, tree, distance.

   (b) What is the shortest path problem ? Write down the Dijkstra's algorithm.

4. (a) Consider A = B = C = R and Let $f : A \to B$ and $g : B \to C$ be defined by $f(x) = x + 9$ and $g(y) = y^2 + 3$. Find the following composition function :
   (i) $f \circ f$
   (ii) $f \circ g$
   (iii) $g \circ f$

   (b) Explain injective, surjective and bijective functions with an example.

5. (a) Consider a relation R defined by X = Y (mode) 5 on set N. Find out the equivalence classes of R.

   (b) Prove Intersection of sets is distributive w. r. t. Union sets.

6. (a) Give difference between equivalence relation and partial order relation using proper example.

   (b) Show that A Subset B implies that union (B – A) = 0.

7. (a) State the modus tollens rule and prove that it is valid.

   (b) What are the planner graphs ?

8. Write short notes on any *four* of the following :
   (i) Propositional logic connectives
   (ii) K-map technique
   (iii) Permutation function
   (iv) Types of relation
   (v) Use of function in coding theory

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Paper Code : 21307

F–407

B. C. A. (Second Semester) EXAMINATION, 2018

(New Course)

Paper No. BCA—N–202

DISCRETE MATHEMATICS

Time : Three Hours] [*Maximum Marks : 70*

Note : Attempt any *five* questions.

1. (a) Show that :

$$p \to q, q \to r \vdash p \to r$$

is valid argument.

(b) Explain the concept of tautology and contradiction using suitable example.

2. (a) How is predicate logic different from propositional logic ?

(b) Draw truth table for :

(i) $P \uparrow Q \uparrow R$

(ii) $\neg (p \wedge \neg q) \to (p \vee \neg r)$

3. (a) What are the logic gates ? Derive AND and OR gate using them/universal gates.

(b) Draw logic/ckt diagram to represent the Boolean expression :

(i) $(x_1 x_2)' + (x_1' x_2')$

(ii) $(x_1 + x_2 + x_3) x_4 + x_5$

(iii) $(x_1')'$

4. (a) What are the normal forms ? Convert the following expression into DNF and CNF :

$$(p \to q) \wedge (\neg p \vee \neg q)$$

(b) What is K-map ? Draw a structure of 4-variable K-map.

5. (a) Write and explain the Dijkstra's algorithm for shortest path problem.

(b) Define the following terms :

(i) Tree and Rooted tree

(ii) Bipartite graph

(iii) Complete graph

(iv) Regular graph

(v) Null graph

(vi) Distance in a graph

(vii) Path in a graph

6. (a) Let f: be a function defined as $f(x) = 2x + 1$ and g : be a function defined as $g(x) = (x+1)^2$. Determine the following :

(i) $f \circ g$

(ii) $g \circ f$

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(b) What are recursively defined functions ? Give suitable example.

7. (a) Let A = {5, 6, 7, 8}. Give example of relations which is :

(i) Reflexive

(ii) Symmetric

(iii) Transitive

(iv) Antisymmetric

(b) What do you mean by term "POSET" ? Describe an equivalence relation.

8. Write short notes on any *four* of the following :

(i) Planer graph

(ii) Graph colouring

(iii) Conditional statements

(iv) Laws of set theory

(v) Injective and bijective function

(vi) Digraph

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